

Optimal precision dependence for geometric Rényi quasi-entropy in the previously open range $1 < \alpha \leq 2$

Candidate full solution for `fa_banach_001`

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Status. Candidate full solution, likely valid; human review recommended. For every fixed $\alpha \in (1, 2]$, the source algorithm’s $\tilde{O}_\alpha(1/\varepsilon)$ precision dependence is optimal, even on commuting full-rank qubit states with constant condition bounds. The same example also gives matching precision lower bounds for the logarithmic geometric Rényi entropy and a quadratic copy lower bound.

1 Source problem

Liu, Wang, Wilde, and Zhang define the geometric Rényi relative quasi-entropy of positive definite density matrices by

$$\hat{F}_\alpha(\rho, \sigma) = \text{Tr}(\sigma \#_\alpha \rho) = \text{Tr} \left[\sigma (\sigma^{-1/2} \rho \sigma^{-1/2})^\alpha \right], \quad \alpha \in (0, 1) \cup (1, 2]. \quad (1)$$

They construct an additive- ε estimator using $\tilde{O}_\alpha(1/\varepsilon)$ purification-oracle queries when the lower spectral bounds are fixed. Their Lemma 20 proves a matching $\Omega_\alpha(1/\varepsilon)$ lower bound for $0 < \alpha < 1$. Immediately afterward they ask whether optimality still holds for $1 < \alpha \leq 2$ [1, PDF p. 24, after Lemma 20].

to $\mathcal{O}_\rho$ and $\mathcal{O}_\sigma$, where $\kappa_\rho, \kappa_\sigma > 0$ satisfy $\rho \geq I/\kappa_\rho$ and $\sigma \geq I/\kappa_\sigma$.

Proof. See Appendix H. □

Notably, we show that our quantum algorithm for estimating the geometric fidelity $\hat{F}_{1/2}(\rho, \sigma)$ achieves an optimal ε -dependence. The optimality also holds for $\hat{F}_\alpha(\rho, \sigma)$ with $\alpha \in (0, 1)$.

Lemma 20 (Optimal ε -dependence of geometric α -fidelity estimation). *Suppose that $\mathcal{O}_\rho$ and $\mathcal{O}_\sigma$ prepare purifications of mixed quantum states ρ and σ , respectively, with $\rho \geq I/\kappa_\rho$ and $\sigma \geq I/\kappa_\sigma$, where $\kappa_\rho, \kappa_\sigma > 0$. Then, for any constant $\alpha \in (0, 1)$, any quantum algorithm that estimates $\hat{F}_\alpha(\rho, \sigma)$ within additive error ε requires query complexity $\Omega(1/\varepsilon)$ even if $\kappa_\rho = \kappa_\sigma = \Theta(1)$, where $\Omega(\cdot)$ hides a constant factor that depends only on α .*

Proof. See Appendix I. □

It remains an open problem for optimality still holds if $\alpha \in (1, 2]$. However, note that when $\alpha \in (1, 2]$, the inequality $\hat{F}_\alpha(\rho, \sigma) \geq 1$ holds, and so $\hat{F}_\alpha(\rho, \sigma)$ cannot be interpreted as a fidelity for these values of α ; thus, different techniques are required in order to establish optimality.

Remark 4. *Similar to Remark 3, for estimating the corresponding quantities, we can extend Theorems 18 and 19 to quantum algorithms with sample complexity $\text{poly}(\kappa_\rho, \kappa_\sigma) \cdot \tilde{O}(1/\varepsilon^3)$, and extend Lemma 20 to a sample lower bound of $\Omega(1/\varepsilon^2)$.*

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Figure 1: The source’s lower bound for $0 < \alpha < 1$ and the explicit open problem for $1 < \alpha \leq 2$ (arXiv:2405.00673, PDF p. 24).

Transcription. “It remains an open problem [whether] optimality still holds if $\alpha \in (1, 2]$.” The source notes that $\widehat{F}_\alpha \geq 1$ in this range and anticipates that different techniques may be required.

2 Idea of the proof

No new noncommutative construction is needed. Use the source’s own commuting Bernoulli hard pair and inspect the scalar target as the Bernoulli bias varies. The first derivative is nonzero on both sides of $\alpha = 1$. It changes sign when one keeps the source’s ordering of the two hard instances, but query discrimination only uses the absolute output gap. Reversing the decision threshold preserves a linear gap while the purification oracles remain only $O(t)$ apart in Hellinger distance. Belovs’ oracle-discrimination theorem then forces $\Omega(1/t)$ queries. Taking $t \asymp_\alpha \varepsilon$ proves the result.

3 Query lower bound

For $t \in [0, 1/4]$, put

$$\rho_t = \begin{pmatrix} (1+t)/2 & 0 \\ 0 & (1-t)/2 \end{pmatrix}, \quad \eta = \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}. \quad (2)$$

Both ρ_t and η dominate $I/4$. Since they commute,

$$\begin{aligned} g_\alpha(t) &:= \widehat{F}_\alpha(\rho_t, \eta) \\ &= \left(\frac{1}{4}\right)^{1-\alpha} \left(\frac{1+t}{2}\right)^\alpha + \left(\frac{3}{4}\right)^{1-\alpha} \left(\frac{1-t}{2}\right)^\alpha. \end{aligned} \quad (3)$$

Lemma 1 (Linear separation). *For every fixed $\alpha > 1$, there are constants $c_\alpha, t_\alpha > 0$ such that*

$$g_\alpha(2t) - g_\alpha(t) \geq c_\alpha t \quad (0 < t \leq t_\alpha).$$

The same assertion holds with g_α replaced by $r_\alpha = (\alpha - 1)^{-1} \log g_\alpha$.

Proof. Differentiation at zero gives

$$g'_\alpha(0) = \alpha 2^{-\alpha} \left[\left(\frac{1}{4}\right)^{1-\alpha} - \left(\frac{3}{4}\right)^{1-\alpha} \right] > 0, \quad (4)$$

because $1 - \alpha < 0$. By continuity, $g'_\alpha(u)$ is bounded below by $g'_\alpha(0)/2$ on a sufficiently short interval $[0, 2t_\alpha]$. Integrating over $[t, 2t]$ proves the first assertion. Moreover,

$$r'_\alpha(0) = \frac{g'_\alpha(0)}{(\alpha - 1)g_\alpha(0)} > 0,$$

so the identical continuity argument proves the second. $\square$

We use the following query lower bound in precisely the oracle model used by the source. If p, q are classical distributions and unitary oracles prepare purifications with amplitudes $\sqrt{p_j}$ and $\sqrt{q_j}$, respectively, then distinguishing the two oracles with constant success probability takes $\Omega(1/H(p, q))$ queries, where

$$H(p, q)^2 = \frac{1}{2} \sum_j (\sqrt{p_j} - \sqrt{q_j})^2.$$

This is Belovs’ distribution-discrimination theorem [2, Theorem 4], also quoted as the lower-bound input in the source.

Theorem 2 (Resolution of the open problem). *Fix $\alpha \in (1, 2]$. Any quantum algorithm which, given purification oracles for ρ and σ satisfying $\rho, \sigma \geq I/4$, estimates $\widehat{F}_\alpha(\rho, \sigma)$ to additive error ε with success probability at least $2/3$ requires*

$$\Omega_\alpha(1/\varepsilon)$$

queries as $\varepsilon \downarrow 0$.

Proof. Let $p_t = ((1+t)/2, (1-t)/2)$, the eigenvalue distribution of ρ_t . For $0 < t \leq 1/4$, every coordinate of both p_t and p_{2t} is at least $1/4$. Hence

$$H(p_t, p_{2t})^2 = \frac{1}{2} \sum_{j=0}^1 \frac{(p_t(j) - p_{2t}(j))^2}{(\sqrt{p_t(j)} + \sqrt{p_{2t}(j)})^2} \leq \frac{t^2}{4},$$

and in particular $H(p_t, p_{2t}) \leq t$.

Let an estimator as in the theorem have additive error ε . By Lemma 1, for all sufficiently small ε we may choose $t = C_\alpha \varepsilon$ so that $t \leq \min(t_\alpha, 1/4)$ and

$$g_\alpha(2t) - g_\alpha(t) \geq 4\varepsilon.$$

Give the estimator either a purification oracle for ρ_t or one for ρ_{2t} as its first input, and give it the fixed purification oracle for η as its second input. Comparing the output with the midpoint of $g_\alpha(t)$ and $g_\alpha(2t)$ distinguishes the two cases whenever the estimation succeeds. Belovs' theorem therefore forces

$$\Omega(1/H(p_t, p_{2t})) = \Omega(1/t) = \Omega_\alpha(1/\varepsilon)$$

queries to the variable oracle, and hence at least that many total queries. All input states dominate $I/4$, as required. $\square$

Combining Theorem 2 with the source upper bound proves that, for every fixed $\alpha \in (1, 2]$, the precision dependence is $\tilde{\Theta}_\alpha(1/\varepsilon)$. The endpoint $\alpha = 2$ is included. At the excluded point $\alpha = 1$, $\hat{F}_1(\rho, \sigma) = \text{Tr } \rho = 1$, so the change at $\alpha = 1$ is genuine rather than a defect of the construction.

4 Two strengthenings

The geometric Rényi entropy is

$$\hat{D}_\alpha(\rho \parallel \sigma) = \frac{1}{\alpha - 1} \log \hat{F}_\alpha(\rho, \sigma).$$

The second part of Lemma 1, followed verbatim by the proof of Theorem 2, gives the following.

Corollary 3 (Entropy precision). *For each fixed $\alpha \in (1, 2]$, additive- ε estimation of $\hat{D}_\alpha(\rho \parallel \sigma)$ from purification oracles requires $\Omega_\alpha(1/\varepsilon)$ queries even for commuting qubit states dominating $I/4$. Thus the source's $\tilde{O}_\alpha(1/\varepsilon)$ entropy algorithm also has optimal precision dependence.*

Corollary 4 (Copy lower bounds). *For either $\hat{F}_\alpha$ or $\hat{D}_\alpha$, estimation to additive error ε from copies of the input states requires $\Omega_\alpha(1/\varepsilon^2)$ copies, under the same fixed- α and constant-conditioning assumptions.*

Proof. For product distributions,

$$H(p_t^{\otimes m}, p_{2t}^{\otimes m})^2 = 1 - (1 - H(p_t, p_{2t}))^m \leq mH(p_t, p_{2t})^2 \leq mt^2/4.$$

Total variation is at most $\sqrt{2}$ times Hellinger distance. Therefore a constant-bias distinction requires $m = \Omega(1/t^2)$. The density matrices commute, so even an arbitrary joint quantum measurement on their tensor powers has exactly the corresponding classical trace-distance limitation. Taking $t \asymp_\alpha \varepsilon$ as above proves the claim; copies of the fixed state η convey no information about the case. $\square$

5 Verification, novelty, and limitations

The derivative, sign, entropy derivative, finite gaps, and Hellinger estimate were checked numerically for $\alpha = 1.01, 1.1, 1.25, 1.5, 1.75, 2$ and four scales 10^{-2} through 10^{-5} by `code/check_hard_instance.py`; all checks passed. This finite computation is only a sanity check, not part of the proof.

The novelty search covered the run registry and all cheap solution/attempt indexes; exact searches for the source's open sentence; searches combining “geometric Rényi relative quasi-entropy”, $\alpha > 1$, quantum query complexity, and precision lower bounds; the current arXiv record; and the 2025 npj Quantum Information publication. Both current source versions still state the problem as open, and no later answer was found through 17 August 2026. The result is elementary and very close to the source proof, so expert review should focus on the purification-oracle convention and on confirming that the sign reversal was not handled under different notation elsewhere.

The theorem concerns precision dependence for fixed α and constant spectral lower bounds. It does not optimize dependence on α , dimension, rank, or condition numbers, and it does not close the gap between the $\Omega(1/\varepsilon^2)$ copy lower bound and the source's $\tilde{O}(1/\varepsilon^3)$ copy upper bound.

References

- [1] N. Liu, Q. Wang, M. M. Wilde, and Z. Zhang, *Quantum algorithms for matrix geometric means*, arXiv:2405.00673; npj Quantum Information **11** (2025), Article 101.
- [2] A. Belovs, *Quantum algorithms for classical probability distributions*, Proceedings of the 27th Annual European Symposium on Algorithms (ESA 2019), LIPIcs 144, Article 16, 2019.